

Antonin Carette

Senior Software Engineer

Education

Diplomas and Teaching

- 2026 **Online Diploma**, Coursera - Imperial College London.
Mathematics for Machine Learning - Verification ID *DYAR9DU3L1BE*
- 2017 **Teaching Assistant**, University of Luxembourg, Luxembourg.
Data Science and Mathematics courses for bachelor students.
- 2014–2016 **Master Degree**, University of Lille, France.
Computer Science, 'Optimization Algorithms and Artificial Intelligence'.
Magna cum laude.
- 2012–2014 **Bachelor Degree**, University of Lille, France.
Computer Science.
Cum laude.

Work Experience

Industry

- 2025–... **Senior Software Engineer**, Tadaweb, Luxembourg.
Building a fast, reliable, and scalable data pipeline. Building an internal Ontology in order to structure data around Tadaweb business.
- 2025 **Senior Software Systems Architect**, Thales Alenia Space, Luxembourg.
Worked on a Rust SDK to develop and execute safe and fast applications on satellites. Investigated system safety constraints, and evaluated WebAssembly as an architecture target for software applications on satellites. Worked on FPGA and ARM processing units.
Programming environment: C/C++, Arm assembly, Rust.
- 2024–2025 **Head of Research & Development**, Red Art Games, France.
Led a team of game developers to build emulators and port old games (MS-Dos games) for modern video game consoles.
- 2022–2025 **Senior Software Developer**, Red Art Games, France.
Made game engine ports (V3X, Adventure Game Studio, Heaps, ...) from PC to modern video game consoles (Sony PlayStation 4 & PlayStation 5, Microsoft Xbox One & Series, and Nintendo Switch). Strong reverse engineering skills, knowledge of (de)compilers and low-level hardware and software optimization skills.
Programming environment: C 11, C++ 14/17, ARM assembly for Nintendo Switch, use of native graphics APIs (DirectX 12, Vulkan, NVN, GNM, GNMP).
- 2021 **R&D Engineer**, DataThings, Luxembourg.
Developed a proof-of-concept and Machine Learning solution to segment and classify fashion clothes.
Programming environment: Python.

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- 2018–2021 **Software Developer, Tadaweb, Luxembourg.**
Developed custom Machine Learning solutions for the business, from proof of concept to ready-to-production products. Production-level code to be deployed on Tadaweb's cloud stack, using SCRUM principles. Helped to benchmark AI software products from famous cloud providers (Microsoft Azure, Google Cloud, etc.).
Programming environment: C/C++, Go.
- 2016–2018 **Data Scientist and Software Developer, DernierCri, France.**
Led Data Science activities for SMEs, and developed back-end services for startups.
Programming environment: Python, Rust, C/C++.
- Research**
- 2016–2017 **Assess and evaluating the energy consumption of Android apps code smells, LATECE team, UQÀM, Montréal.**
Built a strong protocol to measure the energy consumption of Android devices, to assess the impact of code smells on smartphones and Real-Time Operating-Systems (RTOS). The research paper has been submitted and accepted at **SANER 2017**.
Programming environment: Python, Rust, Java (for Android).
- 2015 **Prediction of bugs propagation for big Java projects, SequeL team, INRIA Lille, France.**
Developed an automated method to study and predict the impacts of software bugs in popular Java projects. The research paper has been submitted and accepted at **RAISE 2016**.
Programming environment: Python.
- 2014 **Visualization of V(D)J clones in the case of acute lymphoblastic leukemia decease, Bonsai team, CRISAL Lille, France.**
Developed a visualization tool inside the Vidjil program to visualize V(D)J clones according to their edit distance, in the case of acute lymphoblastic leukemia decease.
Programming environment: JavaScript.

Publications

- 2017 **Mastering Rust: Advanced concurrency, macros, and safe database, Packt Edition**, by Vesa Kaihlavirta.
Lead Reviewer and writing contributor.
- 2017 **Investigating the energy impact of Android smells, SANER 2017**, by Antonin Carette, Mehdi Adel Ait Younes, Geoffrey Hecht, Naouel Moha, and Romain Rouvoy.
First author.
- 2016 **A Learning Algorithm for Change Impact Prediction: Experimentation on 7 Java Applications, RAISE 2016**, by Vincenzo Musco, Antonin Carette, Martin Monperrus, and Philippe Preux.
Second author.

Skills

- **Project Leadership:** Proven ability to lead complex technical initiatives from proof-of-concept (POC) to ready-to-production (RTP) software within cross-functional teams.
- **Applied Problem Solving:** Strong analytical capabilities focused on translating theoretical or research challenges into concrete, real-world engineering solutions.

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- **Technical Communication:** Excellent time management and cross-disciplinary communication skills, ensuring seamless alignment between engineering and research goals.
- **Open-Source Collaboration:** High-level programming and system architecture expertise honed through active, long-term contributions to major open-source ecosystems.

IT Package

- **Core Languages:** Advanced proficiency in solving complex engineering problems using Go, Rust, C/C++, and Python.
- **Machine Learning & AI:** Demonstrated expertise in designing, implementing, and deploying custom AI solutions across supervised, unsupervised, and reinforcement learning paradigms, as well as multi-agent systems.
- **Performance Engineering:** Deep expertise in CPU/GPU debugging and software profiling to identify bottlenecks, reduce memory footprints, and maximize execution speed.
- **Infrastructure & DevOps:** Comprehensive experience managing the full software lifecycle, including Docker containerization, cloud environment deployments, and GitLab CI/CD pipelines.
- **Distributed Systems:** Strong working knowledge of asynchronous architecture and message broker tools, such as RabbitMQ.

Open-Source contributions

- **Machine Learning & AI:** scikit-learn documentation, statsmodel.
- **The Rust community:** cargo-generate, Redox-OS, ar-OS (my own operating system written in Rust), the Rust compiler documentation.
- **Video Games:** ScummVM, SDL2.
- **Other:** FirefoxOS Powertool, SOMCA's Paprika tool, SOMCA's Naga Viper, Calabash.

Languages

French **Mothertongue**
 English **Advanced (C1)**

Personal Interests

- Video game engineering
- Doom 3 (I can easily hold a conversation about Id Software and the development of Doom, Doom 3, and Quake for, at least, three decades)
- Photography, and cinematography
- Fencing